JIM SONG

Software engineer | Full-stack developer

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SUMMARY

- Full Stack Engineer at BNY focused on modernizing banking platforms across loan servicing, cash reconciliation, and payment processing. Responsibilities span system design, application development, legacy migration, performance optimization, and production readiness.
- Take ownership of key technical decisions and delivery across the application stack, translating complex banking requirements into data models, service interfaces, and end-to-end workflows. Provide technical guidance through design discussions and code reviews, supporting engineers across experience levels.
- Led the modernization of legacy cash matching applications, migrating .NET services and stored-procedure-based business logic to Java and Spring Boot microservices. Improved the production auto cash matching rate from 78% to 91–92% through algorithm enhancements, business rule refinement, and real-time data integration.
- Designed and delivered core payment capabilities for BNY's Trade Modernization project, replacing a fully manual legacy payment workflow with automated processing currently observing approximately 97% STP in production. Designed support for underpayment and overpayment scenarios, duplicate-payment safeguards, approvals, and asynchronous payment status handling.
- Build reliable integrations across banking systems, with an emphasis on exception handling, retry behavior, unknown-status reconciliation, and operational visibility. Coordinate with account setup, payment, and other dependent teams to resolve data gaps and integration issues affecting end-to-end processing.
- Optimize data-intensive banking workflows through SQL tuning, in-memory computation, parallel processing, and high-volume Kafka consumption. Reduced a large-scale legacy data migration process from more than 48 hours to less than 24 hours through data analysis and SQL optimization.
- Contribute to production readiness and ongoing support through automated testing, CI/CD, multi-data-center deployment and failover planning, and application monitoring and alerting. Develop integrated user workflows and operational reporting to support daily banking activities.
- Advance team development practices through AI-assisted engineering with Claude, Codex, and Devin. Created shared MCP integrations and reusable skills adopted by team members to support testing workflows and provide project-specific context to AI tools.

SKILLS, TOOLS & PLATFORMS

Programming & scripting languages

Core Java | J2EE | C++ | Python | JavaScript & TypeScript | HTML & CSS

Frameworks & server technology

Spring Boot & Spring Cloud | Hibernate & Spring Data JPA | RESTful API

Ruby on Rails | Tornado Web Server | AWS & Azure | Hadoop | Linux

Databases & frontend libraries

Oracle | MSSQL | Hazelcast | MongoDB | React | Angular

CI/CD & other tools

Maven | Jenkins | Junit | Git | Jira | Splunk

AI tools

Claude | Codex | Devin

EXPERIENCE

The Bank of New York Mellon Corporation, Pittsburgh, PA.

Nov 2025 – Present

Loans Enablement Platform

Description: The Trade Modernization initiative modernizes CIDD trade processing and replaces manual payment workflows with automated capabilities within BNY's Loans Enablement Platform. Built with Java, Spring Boot, Angular micro-frontends, Oracle, and Snowflake, the platform integrates trade ingestion, settlement, and payment services, emphasizing straight-through processing, reliability, and operational visibility.

- Took primary responsibility for payment subsystem design and implementation, defining database schemas, processing rules, and service interfaces, with continued enhancements following production launch.
- Automated payment generation, enrichment, approval handling, submission, and asynchronous status processing, replacing a fully manual workflow and currently observing approximately 97% STP in production.
- Designed payment data models and lifecycle rules for underpayment, overpayment, and duplicate checks, defining when additional payments may proceed or require intervention.
- Implemented retry processing and duplicate-payment safeguards to support failure recovery while controlling repeated submission risks.
- Designed reconciliation for unknown payment outcomes, verifying external status before determining whether further investigation or retry is appropriate.
- Standardized exception handling across payment workflows and integrations, connecting consistent failure reporting with monitoring and alerting to simplify recovery.
- Coordinated with account setup, payment, and other external teams to align interfaces, resolve missing data dependencies, and support end-to-end processing.
- Reduced large-volume legacy data migration time from more than 48 hours to less than 24 hours through data analysis and SQL optimization, with data cleanup supporting migration readiness.
- Built and optimized a Kafka consumer to handle message volume from a high-traffic, company-wide shared topic.
- Implemented monitoring and alerting with Moogsoft, Cloud Probe, Splunk, and email notifications to support production troubleshooting and service recovery.
- Developed a daily-refresh Power BI report for overdraft accounts to support operational monitoring and follow-up.
- Guided system design and reviewed code across backend, frontend, database, and integration changes, addressing correctness, performance, and maintainability while supporting newer engineers.
- Applied Claude, Codex, and Devin to implementation, code analysis, and unit test generation, evaluating generated changes through engineering review.
- Created team-adopted MCP integrations and reusable skills for testing workflows and project context delivery, standardizing recurring AI-assisted engineering tasks.

Technologies used: Java | Spring Boot | Angular | Micro-frontends | Oracle | Snowflake | Kafka | SQL | Power BI | Moogsoft | Cloud Probe | Splunk | Claude | Codex | Devin | MCP

The Bank of New York Mellon Corporation, Pittsburgh, PA. **Nov 2023 – July 2025**

Loans Enablement Platform

Description: The Loans Enablement Platform modernizes and integrates BNY's legacy loan servicing applications, with a focus on cash reconciliation, automated matching, and a unified user experience. The project replaces .NET services and Microsoft SQL Server stored-procedure-based business logic with Java and Spring Boot microservices, using Hazelcast and Snowflake to support data processing and computation. A shared micro-frontend architecture brings loan servicing workflows into a consistent platform while supporting independent development and deployment across teams.

- Led key aspects of platform architecture and delivery, including technology selection, solution prototyping, service design, and CI/CD setup. Translated legacy application behavior and banking requirements into technical designs and implementation plans.
- Led the migration of cash matching functionality from legacy .NET applications and Microsoft SQL Server stored procedures to Java and Spring Boot microservices. Analyzed existing business logic and dependencies to preserve reconciliation rules while restructuring processing for the new platform.
- Improved the production auto cash matching rate from 78% to 91–92% through matching algorithm enhancements, business rule refinement, and real-time data integration, reducing the proportion of cash items requiring manual reconciliation.
- Optimized computation-intensive matching workflows by analyzing performance bottlenecks, refining processing logic, and applying parallel computation with Hazelcast's in-memory data grid to accelerate large-volume data processing.
- Designed and developed a unified data access layer across multiple data sources, providing shared integration patterns that simplified cross-team collaboration and reduced duplicated data access logic.
- Implemented Angular micro-frontend infrastructure and reusable components for daily loan servicing activities, providing a consistent user experience while enabling participating teams to develop and deploy capabilities independently.
- Developed an AI-based issuer name recognition prototype to evaluate opportunities to improve matching across issuer name variations and address cases not readily resolved by existing matching rules.
- Designed and implemented automated business acceptance tests with Cucumber, UI workflow tests with Selenium, and mutation testing to validate reconciliation behavior and assess test effectiveness. Integrated SonarQube to support ongoing code quality checks.
- Configured OAuth2 authentication for Spring Boot services and designed the platform's cloud gateway and routing strategies to support secure access and integration across applications.
- Established GitLab CI pipelines for automated build, testing, artifact publishing, and deployment across data centers. Applied trunk-based development practices and integrated BNY's internal container platform into the application delivery workflow.
- Prepared a feature-toggle-based failover plan using Togglz for applications deployed across three data centers, defining how application behavior could be controlled during failover.
- Used Splunk for production log analysis and issue investigation, and Grafana dashboards to monitor application health, memory consumption, CPU usage, and I/O performance, supporting troubleshooting and performance analysis.
- Contributed to design discussions and code reviews across services, data processing, and frontend integration, helping team members understand business rules, evaluate implementation choices, and maintain consistency across the platform.

Technologies used: Java 21, Spring Boot, Spring Cloud, Spring Data, Hazelcast, Snowflake, Microsoft SQL Server, Kafka, Angular, Micro-frontends, OAuth2, GitLab CI, Cucumber, Selenium, Mutation Testing, SonarQube, Splunk, Togglz, Grafana

Turner Construction Company, Los Angeles, CA. **Jan 2022 – Oct 2023**

Construction Project Management System (CPMS)

Description: The Construction Project Management System manages our construction projects from preconstruction to closeout. As part of the company's digital transformation process, this system aid in every aspect of the construction process by connecting every project contributor with the owner and contractor. Turner uses this portal to deliver tailored services including budget management, progress monitoring, and documentation. It enables users to communicate across disparate teams, making it easier to work together by eliminating silos and establishing a single source of information.

- Played an integral role in the Analysis, Design, Development, System Testing, and User Acceptance Testing phases of CPMS, aligning closely with Agile Methodology and the SCRUM Cycle model.
- Leveraged Java 8's advanced features, notably Lambda Expressions and Stream API, to enhance backend operations' performance and adaptability.
- Architecturally designed the process engine to facilitate code reusability, enabling tailored workflow applications across diverse projects. Enhanced task workflows via the adept use of Spring IOC and service provider interface (SPI).
- Refined and optimized features within the project management service, emphasizing streamlined data access and operations through Spring Data JPA integrated with MySQL.
- Successfully tackled the challenge of migrating legacy services by seamlessly transitioning to a microservices framework and overseeing efficient batch data migration.
- Implemented a secure service approach to guarantee precise access to key assets. Employed the BASE methodology to ensure eventual consistency, while utilizing CDC and message queues, like Kafka, for handling less critical data sets.
- Led the creation of an interactive frontend dashboard, effectively representing complex project metrics using a blend of HTML5, CSS3, JavaScript, and React technologies.
- Integrated AWS S3 for efficient storage of process diagrams, templates, and other project-centric documents. Actively monitored module activities via CloudWatch and accelerated content delivery using CloudFront.
- Incorporated Log4J to provide comprehensive tracking and monitoring of events, operations, and potential issues within the process engine and project management service.
- Relied on GIT as the cornerstone for version control, ensuring streamlined collaboration throughout the development of these sophisticated modules.
- Embraced a container-centric strategy with Docker, orchestrating robust deployments using Kubernetes. This ensured both scalability and resilience for critical components. Further enhanced the development lifecycle by integrating Skaffold, expediting iterative processes.

Technologies used: Java 8, Spring IOC, Spring MVC, Spring Security, Spring Boot, Spring Cloud, Spring Data JPA, HTML5, CSS3, JavaScript, React, Hibernate, S3, CloudWatch, Restful, MySQL, Log4J, Kubernetes, Docker, GIT

EDUCATION

University of Southern California, Los Angeles, CA, United States
Master of Science in Computer Science (High Performance Computing and Simulations)

Soochow University, Suzhou, Jiangsu, China
Bachelor of Engineering in Computer Science and Technology